ENGINEERING

PORTFOLIO

Of GHALIH AHMAD SALAMUN

+62 8125 1700 170 | salamunghalih@gmail.com| www.linkedin.com/in/ghalih-salamun

ABOUT ME

GHALIH

Robotics and Artificial Intelligence Engineering graduate from Airlangga University, recognized as the Faculty's Most Outstanding Graduate, with hands-on experience in underwater robotics and intelligent robotic systems.

Interested in autonomous underwater systems, robotics, intelligent vehicles, and industrial automation.

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WHAT DO I DO?

UNIVERSITAS AIRLANGGA | Robotics and Artificial Intelligence Engineering Aug 2022 - Jul 2026

Graduated with the Faculty's Most Outstanding Graduate distinction.

MARE VICTOR | Head Mechanic & Senior Mechanic
Nov 2023 - Jul 2026

MARE VICTOR is a team I co-founded to research and produce a competition-ready Autonomous Underwater Vehicle (AUV) and Remote Operated Underwater Vehicle (ROV).

Integrated Mechano-Biosystem Lab, NCKU Taiwan | Research Intern Aug 2025 - Dec 2025

Developed a semi-autonomous robotic microtissue handling system integrating a DOBOT MG400 robotic arm, computer vision, and a custom suction-based end-effector.

SKILLS :

- 3D Computer-Aided Design (CAD)
- Computational Fluid Dynamics (CFD)
- Laser-cutting
- 3d printing
- Electrical design layout
- Python
- Computer Vision
- Sensor Integration
- Mobile Robot
- Control Design

TOOLS :

INTERNSHIP PROJECT

Autonomous Microtissues Manipulating Arm

AUG 2025 - DEC 2025

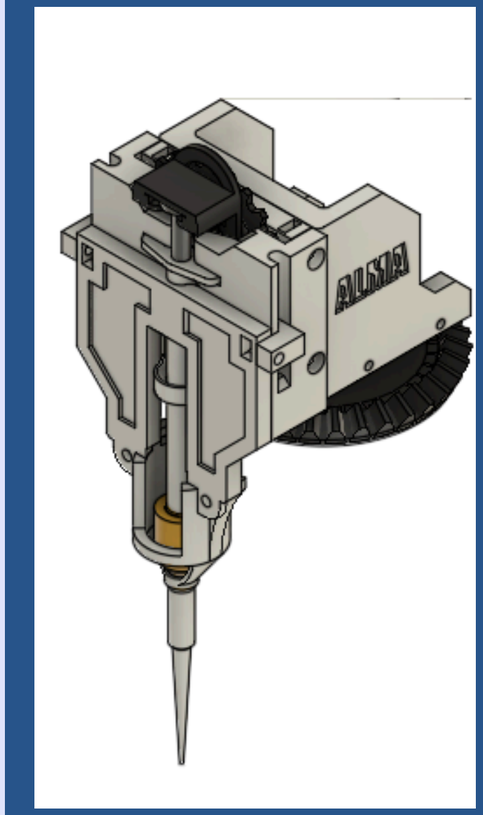
Key Responsibilities :

- Developed a semi-autonomous robotic microtissue handling system integrating a DOBOT MG400 robotic arm, computer vision, and a custom suction-based end-effector.
- Designed and fabricated a 1 cc syringe-based pipette end-effector for controlled pickup and transfer of microtissues between Petri dishes and well plates.
- Implemented YOLOv8-based object detection and homography transformation to identify microtissue positions and convert image coordinates into the robot's workspace.
- Performed system calibration and experimental validation, achieving 0.853 precision, 0.921 recall, 0.886 F1-score, and 0.922 mAP@0.5 for microtissue detection.
- Evaluated robotic positioning accuracy through repeated experiments, achieving a 1.424 mm mean error and 1.881 mm RMS error.

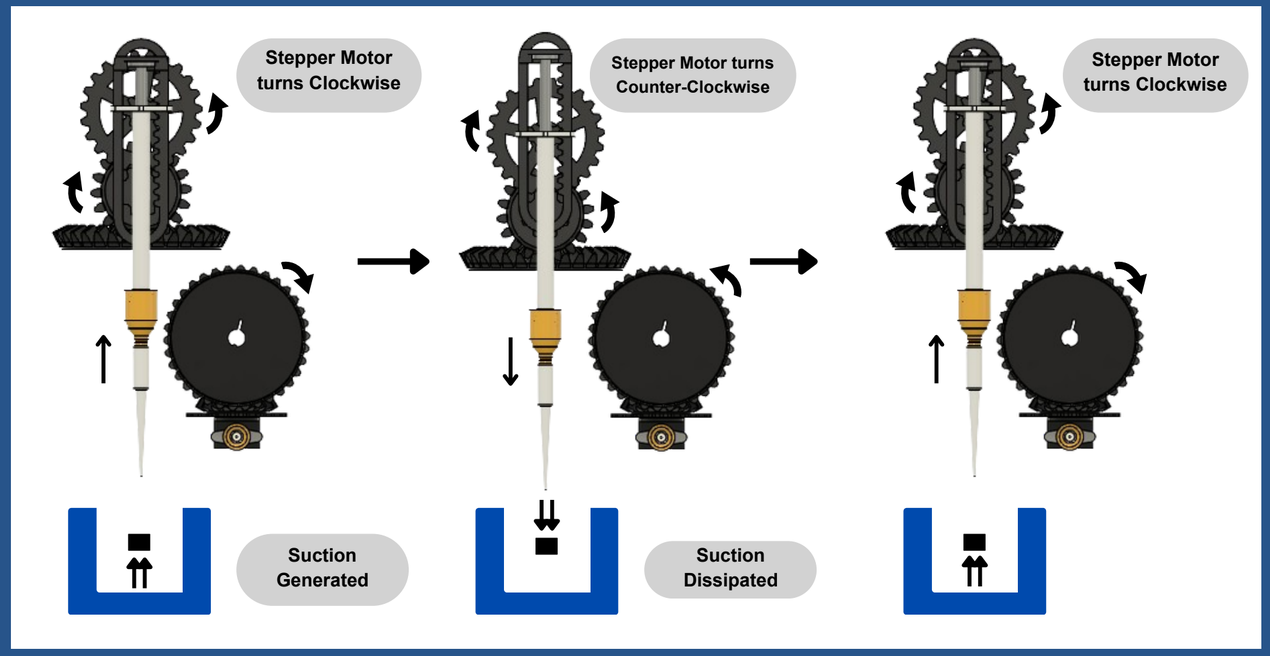
INTERNSHIP PROJECT

Autonomous Microtissues Manipulating Arm

AUG 2025 - DEC 2025



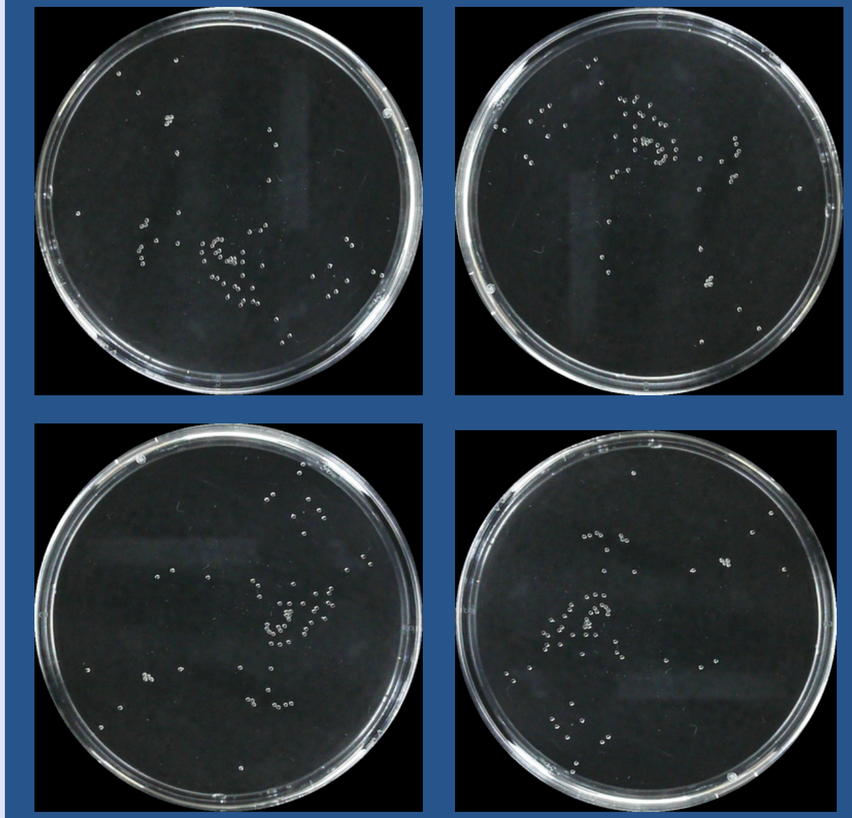
The Custom-designed End-Effector is equipped with a gear transmission system to move the plunger of a commercial 1 cc syringe and generate suction to move microtissues or liquid.

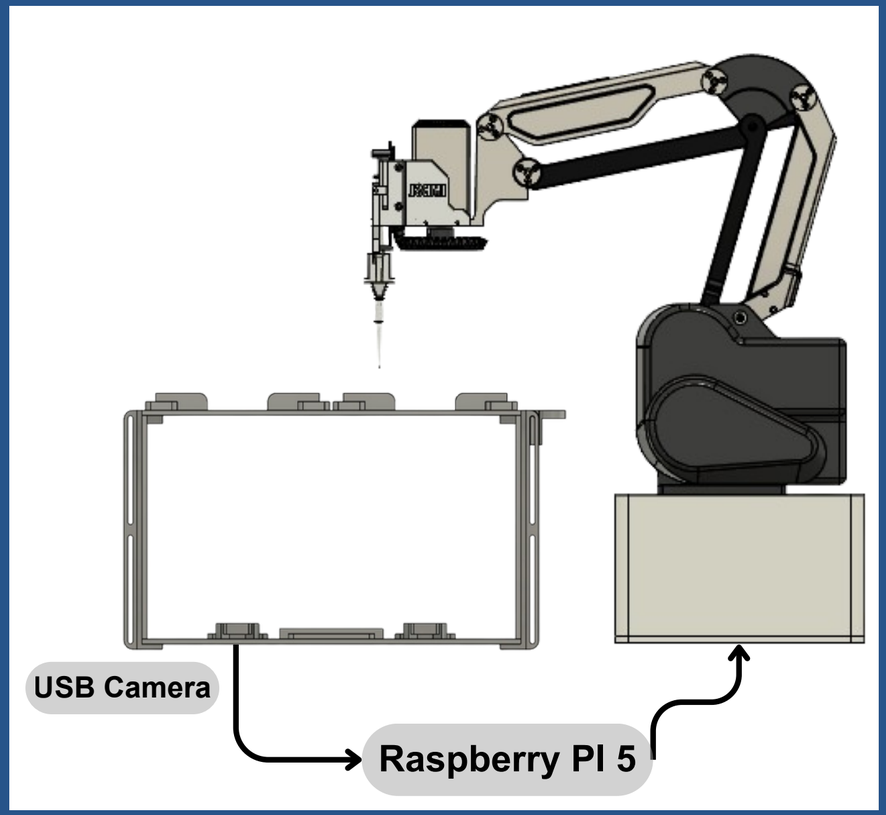


INTERNSHIP PROJECT

Autonomous Microtissues Manipulating Arm

AUG 2025 - DEC 2025





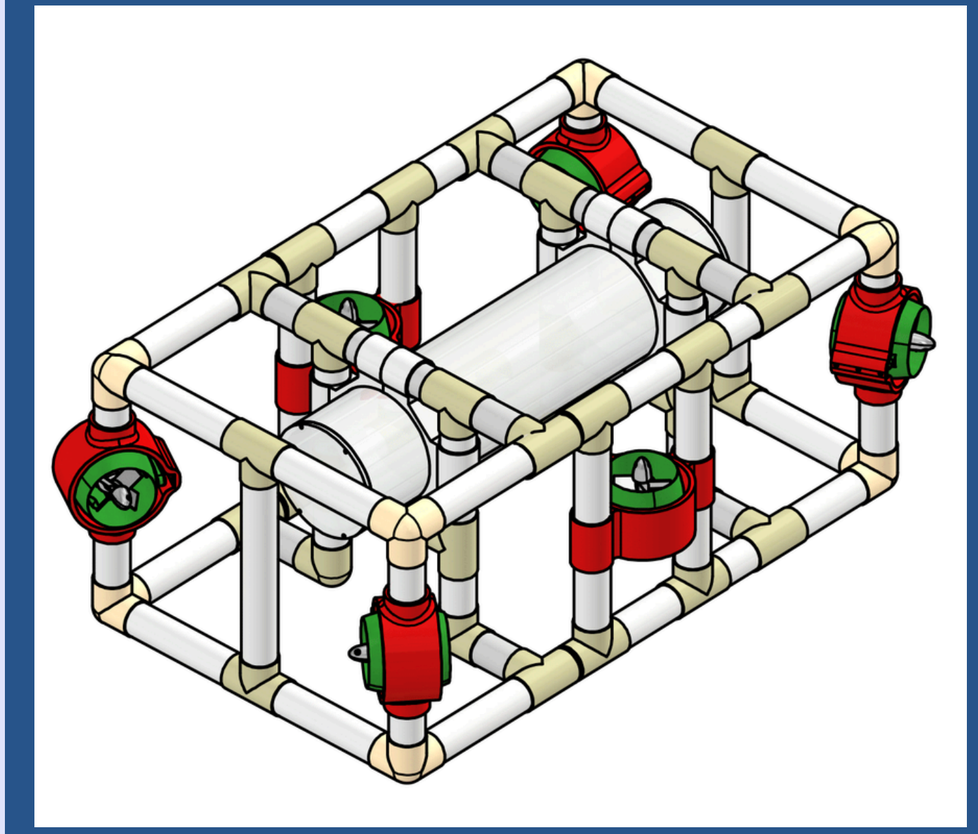
The USB Camera detects Microtissues on the petri dish and The Raspberry Pi 5 locates its positions and move The DOBOT MG400 to its position and move it.

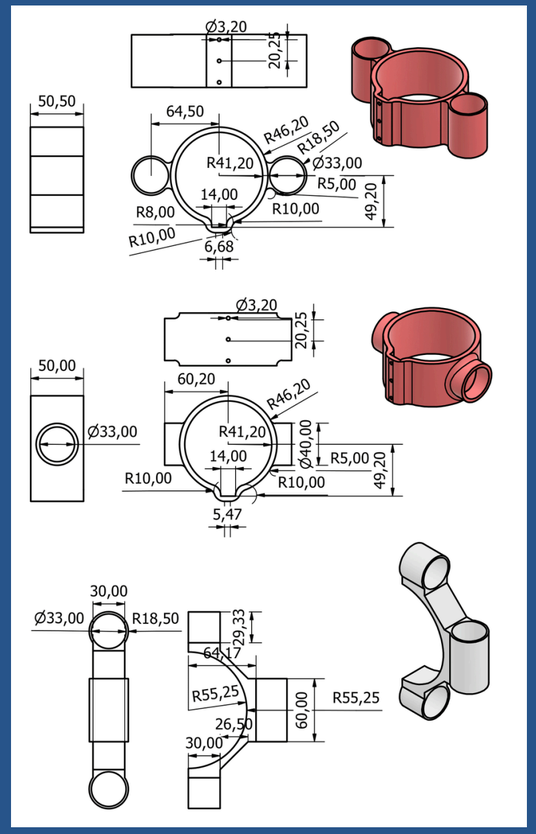
ROV and AUV PROJECTS

NOV 2023 - MAR 2025

MARVIC 1.0

NOV 2023 - MAR 2024

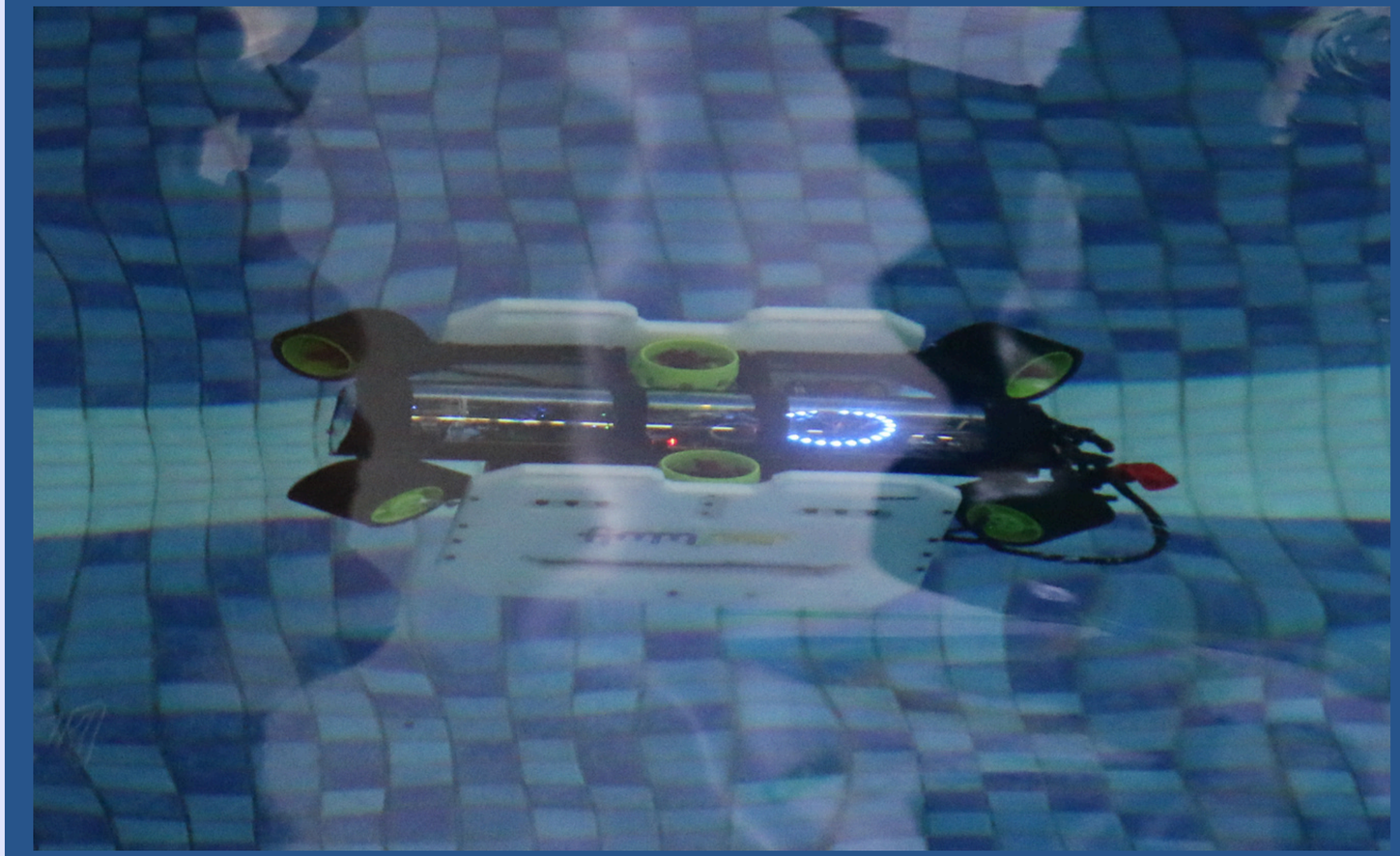




- Designed mechanical parts using Autodesk Inventor for MARVIC's first ROV prototype.
- Fabricated components through laser cutting and 3D printing, ensuring design-to-build continuity.
- Conducted waterproofing validation tests on enclosures and connections to ensure safety.
- Assembled and integrated all mechanical systems, contributing to MARVIC's readiness for KRBAI 2024.

MARVIC 2.0

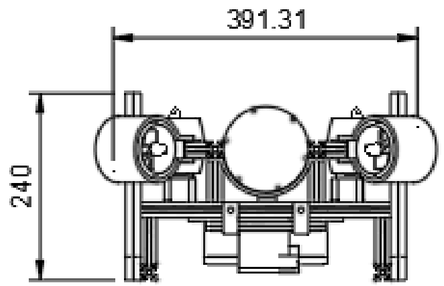
APR 2024 - JUL 2024

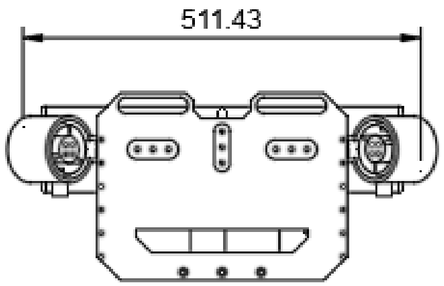


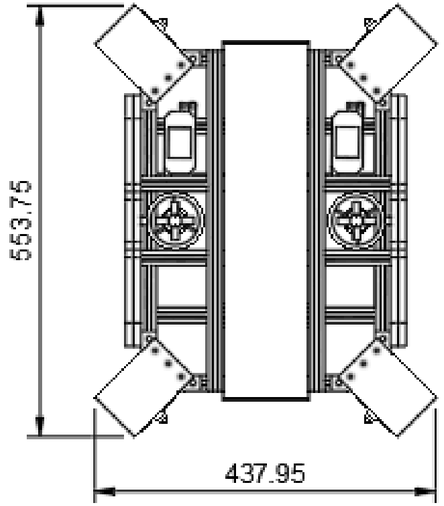
- Design the AUV and mechanism to carry and drop a golf ball throughout obstacles underwater.
- Conducted waterproofing tests on pressure-sensitive components and assisted in full-system integration and final build.
- Fully assembled the AUV with the dimension of 55 x 44 x 24 cm, capable of navigating competition obstacles and gates, completing 3 of 4 tasks in 1 minute 22 seconds.
- Achieved 9th place nationally among 18 participating universities as part of MARE VICTOR.

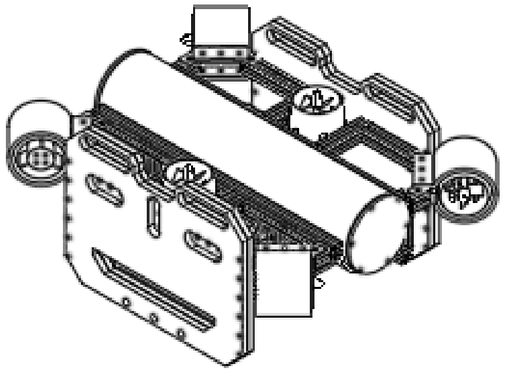
MARVIC 2.0

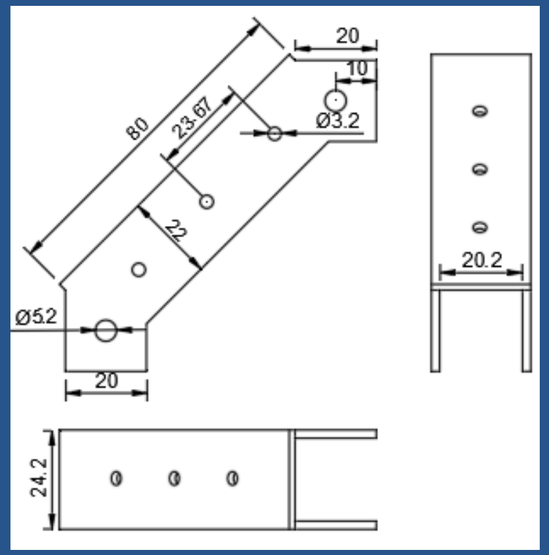
APR 2024 - JUL 2024

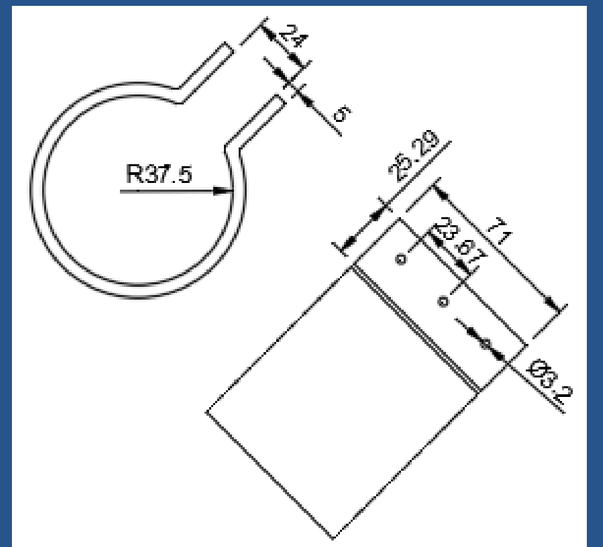


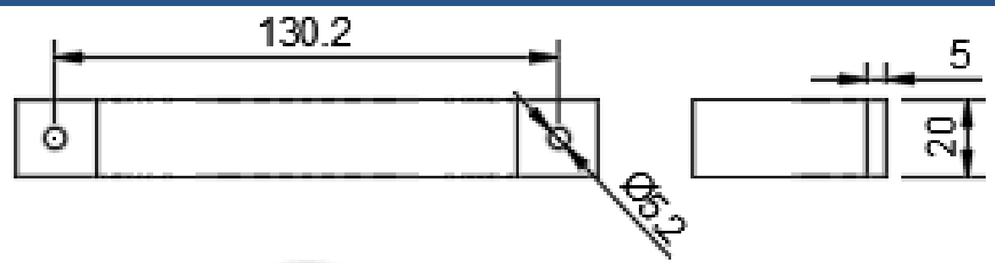


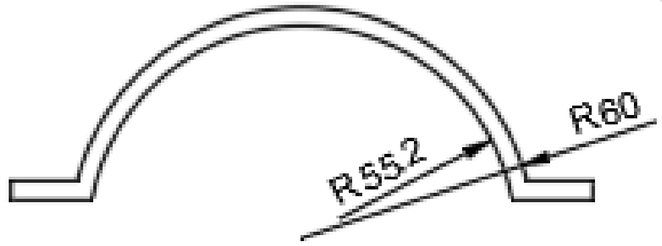






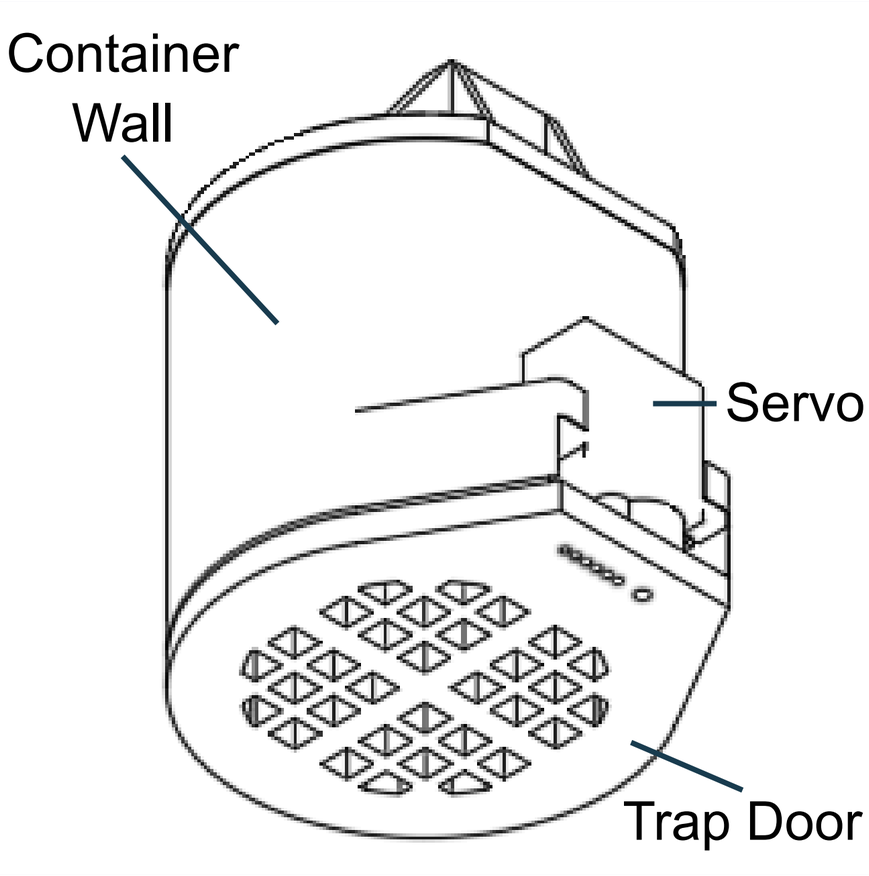


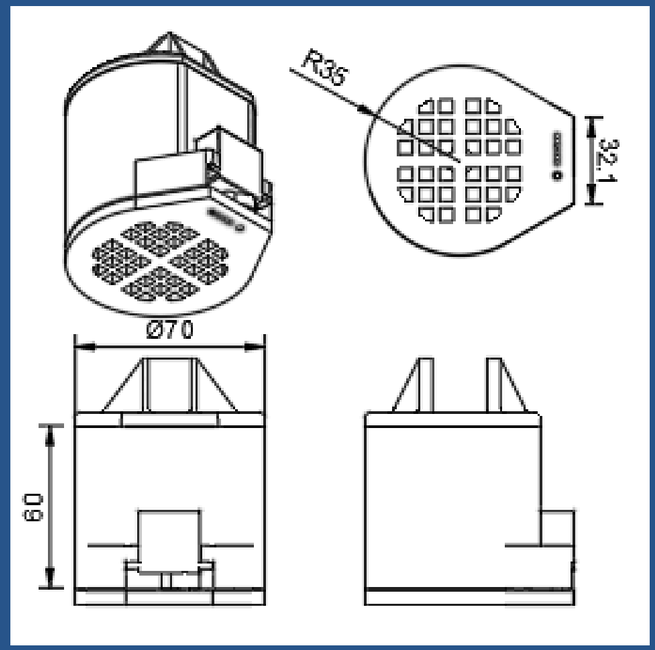




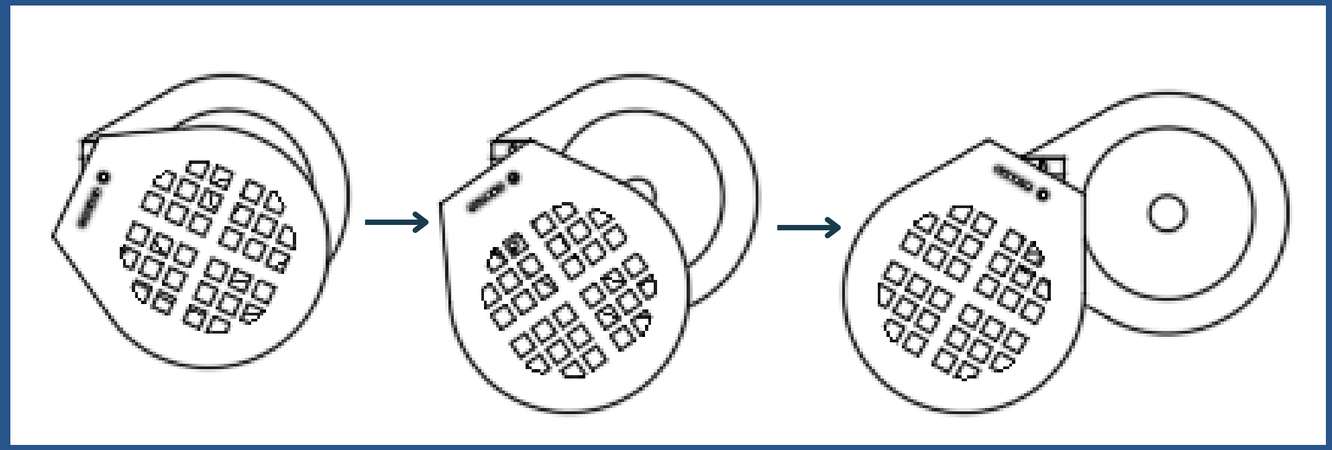
MARVIC 2.0

APR 2024 - JUL 2024





Dropping a Golf Ball Underwater.



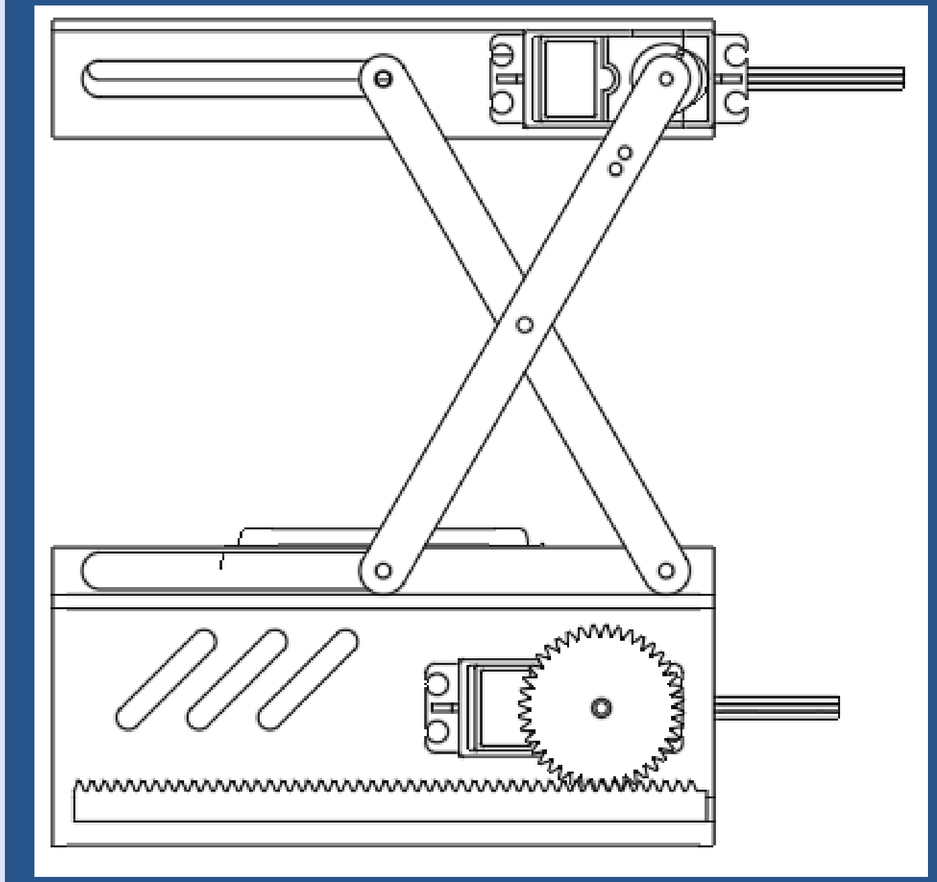
MARVIC 2.5

NOV 2024 - MAR 2025

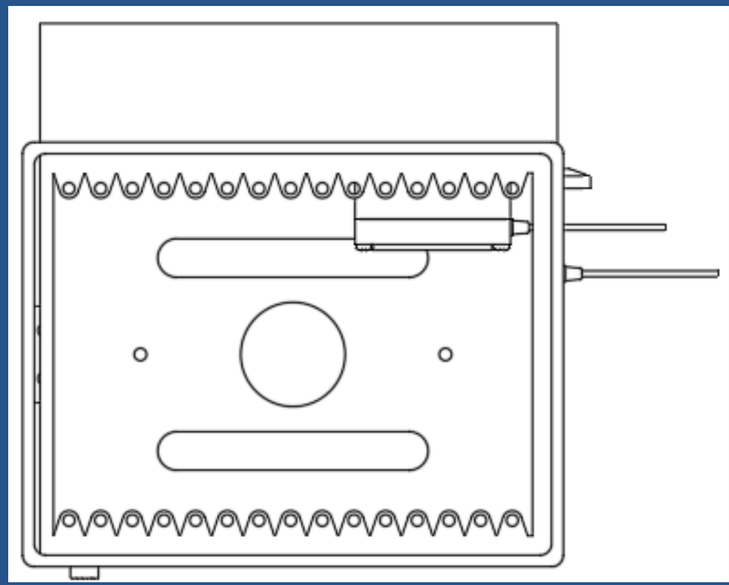
- Drafted mechanical parts and enclosure systems using Autodesk Fusion, ensuring fit, modularity, and buoyancy control.
- Fabricated core components with laser cutting and 3D printing.
- Conducted waterproofing tests and implemented sealing.
- Contributed to a system capable of navigating gates, retrieving golf balls, and performing underwater payload drops in an international competition setting.
- Achieved 21st place among 45 international teams.

MARVIC 2.5

NOV 2024 - MAR 2025



A Servo-powered Scissor Mechanism to Drop and Retrieve Golf Ball.



A second Servo is equipped on the bottom to move the trap door to drop the golf ball.

The Trap Door is equipped with lines made of rubber bands that expand when pressed to a golf ball and traps it.

COMPETITION-BASED PROJECT

PORTABLE WHITE BLOOD CELL LEVELS MONITORING DEVICE ON HIV PATIENTS

APR 2024 - OCT 2024

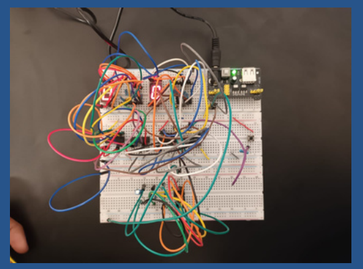
- Designed and prototyped a microfluidic optical biosensor system integrated with IoT for monitoring CD4 white blood cell levels in HIV patients.
- Led the design of a custom camera stand and electrical casing using 3D CAD to meet precise magnification and imaging requirements.
- Conducted hardware research to determine optimal lens-camera configurations for medical image clarity.
- Supported the development of a compact, user-friendly prototype for clinical use by healthcare professionals.
- Presented the research and prototype before a national panel of judges, earning a Bronze Medal in the PKM-Karsa Cipta category.

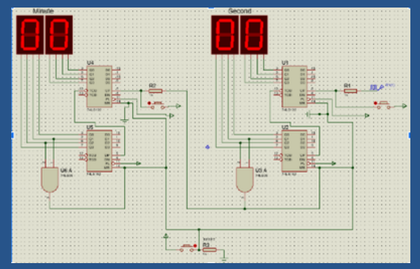
UNI COURSES PROJECTS

Simple Counter-up Timer

MAY - JUNE 2024

This is a project focusing on designing an electrical system of a counter-up timer using ICs only. The ICs we use consist of a counter-up IC 74192, a decoder IC 74LS48 and 4 7-segment display.

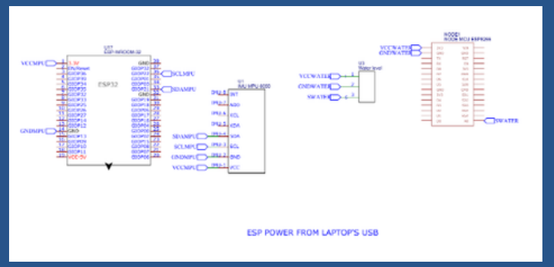




SI-TERA : Prototype of tsunami early detection system

JUNE 2024

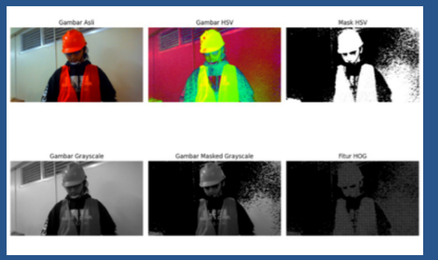
SI-TERA is an IoT project I worked on, focusing on a prototype of early detection of Tsunamis using an Inertial Measurement Unit (IMU) sensor and water sensor.



Personal Protective Equipment (PPE) monitoring program

NOV - DES 2024

This is a computer vision project on making a model to detect PPE use in a workplace environment. We use a Histogram and a Support Vector Machine model to detect PPE usage.

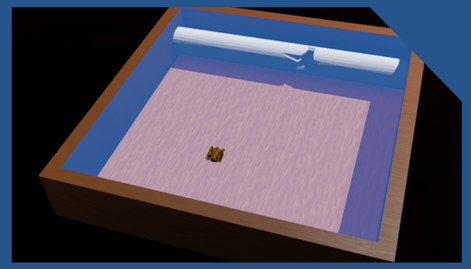


UNI COURSES PROJECTS

Simulation of ROV movement underwater

DES 2024

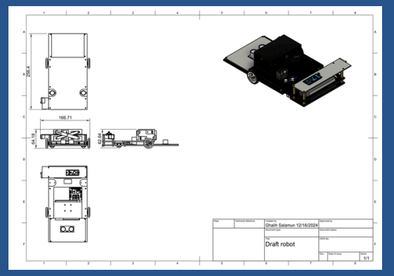
This is a project to learn the fundamentals of mobile robotics using the Webots software. We simulate an underwater environment and the movement of the 6 thrusters ROV to navigate throughout its environment.

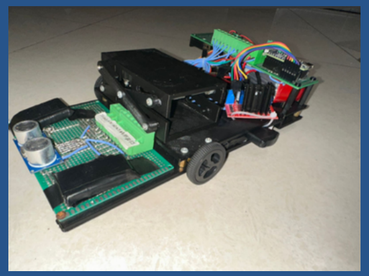


Simple Automated Guided Vehicle (AGV)

AUG 2024 - JAN 2025

Working as the drafter, I designed an AGV to transport mini trolleys using lines and the principle of line following. It operates using an ultrasonic sensor to detect the trolleys.

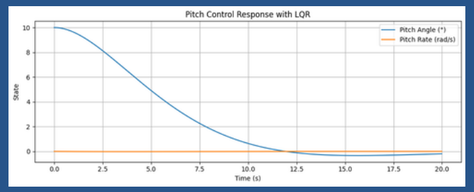




LQR Optimal Control Design to stabilize pitch and heave in AUV

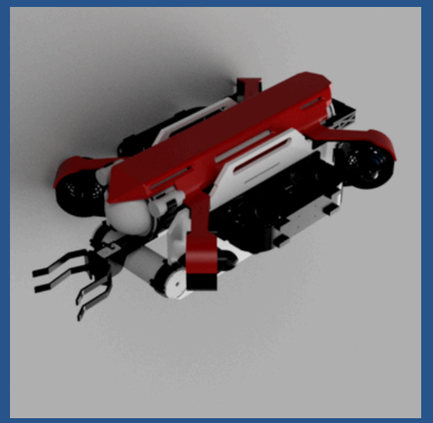
APR - MAY 2025

This project's aim is to utilize the Linear Quadratic Regulator (LQR) method to stabilize the movement of pitch and heave in a Long-range AUV.

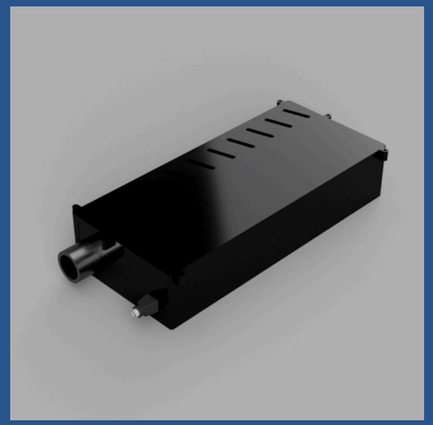


OTHER RELEVANT PROJECTS

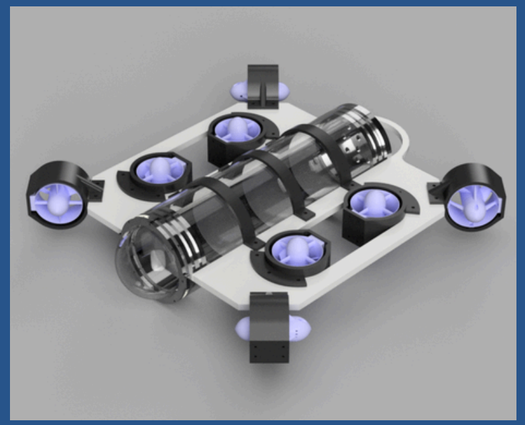
2 Thrusters Long Range AUV for the upcoming study of underwater localization.



8-thruster ROV for planting and monitoring coral reefs. A project shared with students and lecturers at the Bogor Institute of Agriculture. This project has been discontinued.



Electrical Casing for water flow sensor and control. A project shared with students of Kalimantan Technology Institute.



8 Thrusters AUV for monitoring seaweeds and detecting diseases. This project has been discontinued.

ACHIEVEMENTS

Indonesia Underwater Robotics Contest (KRBAI) 2024.

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
PUSAT PRESTASI NASIONAL
BALAI PENGEMBANGAN TALENTA INDONESIA

Sertifikat

Nomor **18754/BPTI/DIKTI/2024**

Kementerian Pendidikan,Kebudayaan,Riset,danTeknologi,Pusat Prestasi Nasional,
Balai PengembanganTalentaIndonesia menyampaikanpenghargaan kepada:

GHALIH AHMAD SALAMUN

Universitas Airlangga
Anggota-MARVIC
atas partisipasi dan pencapaiannya sebagai
FINALIS Divisi **Kontes Robot Bawah Air Indonesia**

pada **Kontes Robot Indonesia (KRI) Tingkat Nasional Tahun 2024**
yang dilaksanakan oleh Balai Pengembangan Talenta Indonesia sebagai unit pelaksana teknis Pusat Prestasi Nasional, Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi di Universitas Muhammadiyah Surakarta, Provinsi Jawa Tengah pada tanggal 1 s.d. 6 Juli 2024.

Jakarta, 5 Juli 2024
Plt. Kepala Balai Pengembangan Talenta Indonesia,

624594223505

Dr. Maria Veronica Irene Herdjiono, S.E., M.Si.
NIP 198103292012122001

Bronze Medalist - 37th National Student Scientific Week (PIMNAS) 2024.

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
PUSAT PRESTASI NASIONAL
BALAI PENGEMBANGAN TALENTA INDONESIA

Sertifikat

Nomor **29252/PPN/DIKTI/2024**

Kementerian Pendidikan,Kebudayaan,Riset,danTeknologi,Pusat Prestasi Nasional,
Balai PengembanganTalentaIndonesia menyampaikanpenghargaan kepada:

GHALIH AHMAD SALAMUN

Universitas Airlangga
atas prestasinya sebagai
Medali Perunggu kategori **Presentasi**
bidang **PKM-Karsa Cipta**

pada **Pekan Ilmiah Mahasiswa Nasional (PIMNAS) ke 37 Tahun 2024** yang dilaksanakan oleh Balai Pengembangan Talenta Indonesia sebagai unit pelaksana teknis Pusat Prestasi Nasional, Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi secara luring bekerja sama dengan Universitas Airlangga, Provinsi Jawa Timur pada tanggal 14 s.d. 19 Oktober 2024.

Jakarta, 18 Oktober 2024
Kepala Pusat Prestasi Nasional,

6244915967791

Dr. Maria Veronica Irene Herdjiono, S.E., M.Si.
NIP 198103292012122001

ACHIEVEMENTS

Singapore Autonomous Underwater Vehicle Challenge (SAUVC) 2025

IEEE SAUVC OES Singapore Chapter

Certificate of Participation

This is to certify that
Ghalih Ahmad Salamun
of
Airlangga University
was a Member of the team that participated in
The Singapore Autonomous Underwater Vehicle Challenge 2025.

Dated
17 March 2025

Certified list of participants

This is a digital certificate; no signature is required.

Faculty's Most Outstanding Graduate - Universitas Airlangga, August 2026

UNIVERSITAS AIRLANGGA
Excellence with Morality

Piagam Penghargaan

Award Certificate

Diberikan kepada:
Present to
Ghalih Ahmad Salamun
atas pencapaiannya sebagai
for his achievement as

WISUDAWAN BERPRESTASI
outstanding graduate

Fakultas Teknologi Maju dan Multidisiplin
Faculty of Advanced Technology and Multidiscipline
pada Wisuda 263 Universitas Airlangga
at 263 graduation ceremony of Universitas Airlangga

Surabaya, 30 Agustus 2026
Rektor/Rector

Prof. Dr. Muhammad Madyan, SE., M.Si., M.Fin.
NIP 197101041998021001

Let’s Craft The Future Together.

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